

Erdős Problem #233

EP-233

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $d_n = p_{n+1} - p_n$, where p_n is the n th prime. Prove that $\sum_{1 \leq n \leq N} d_n^2 \ll N(\log N)^2$.

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>